

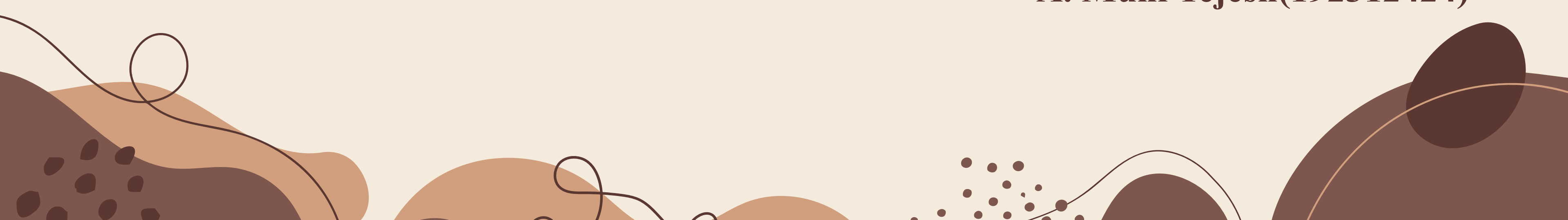


Design and Development of a Blood Donor Management System Using C Programming

**C - PROGRAMMING
(CSA0209)**

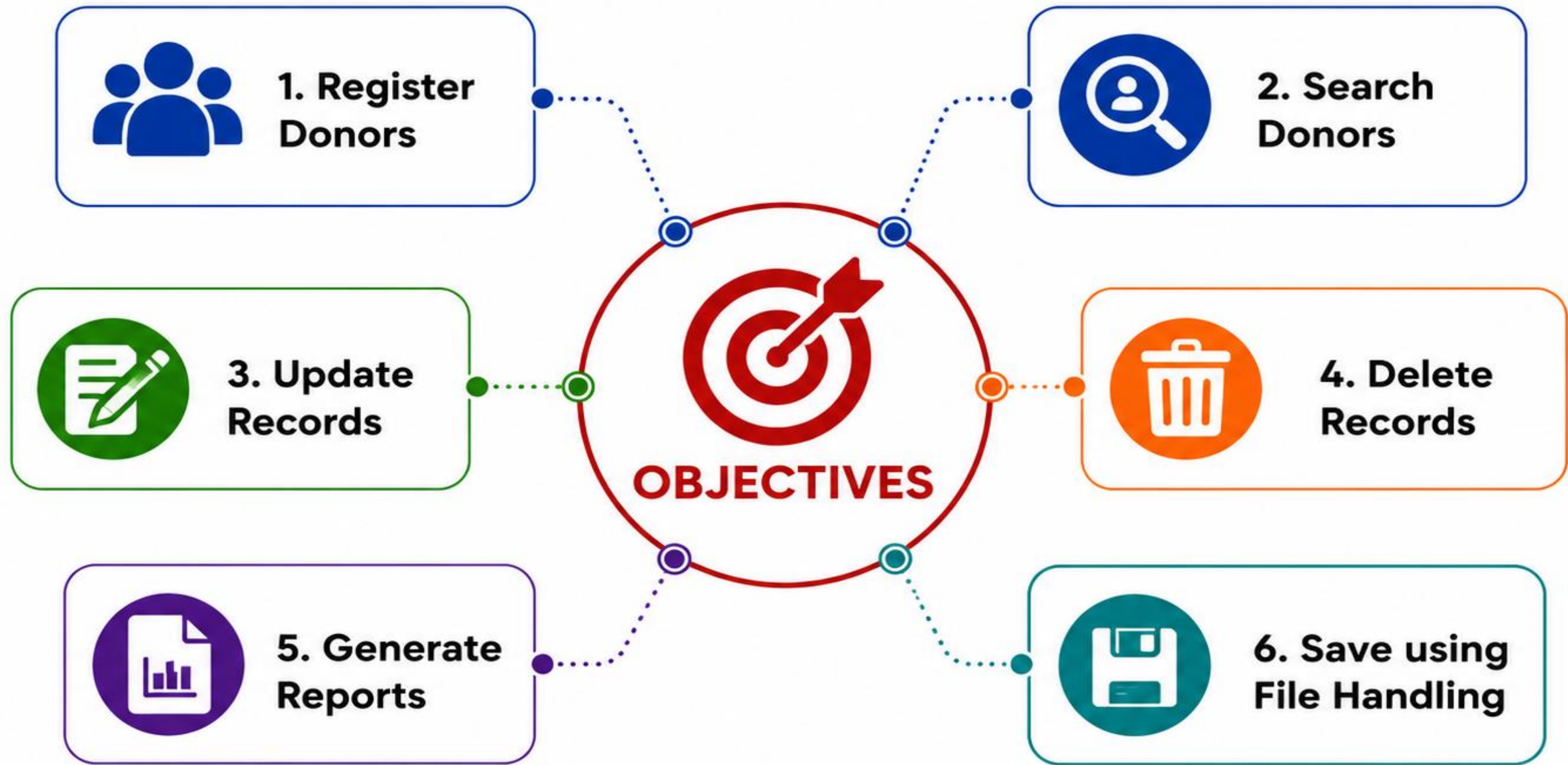
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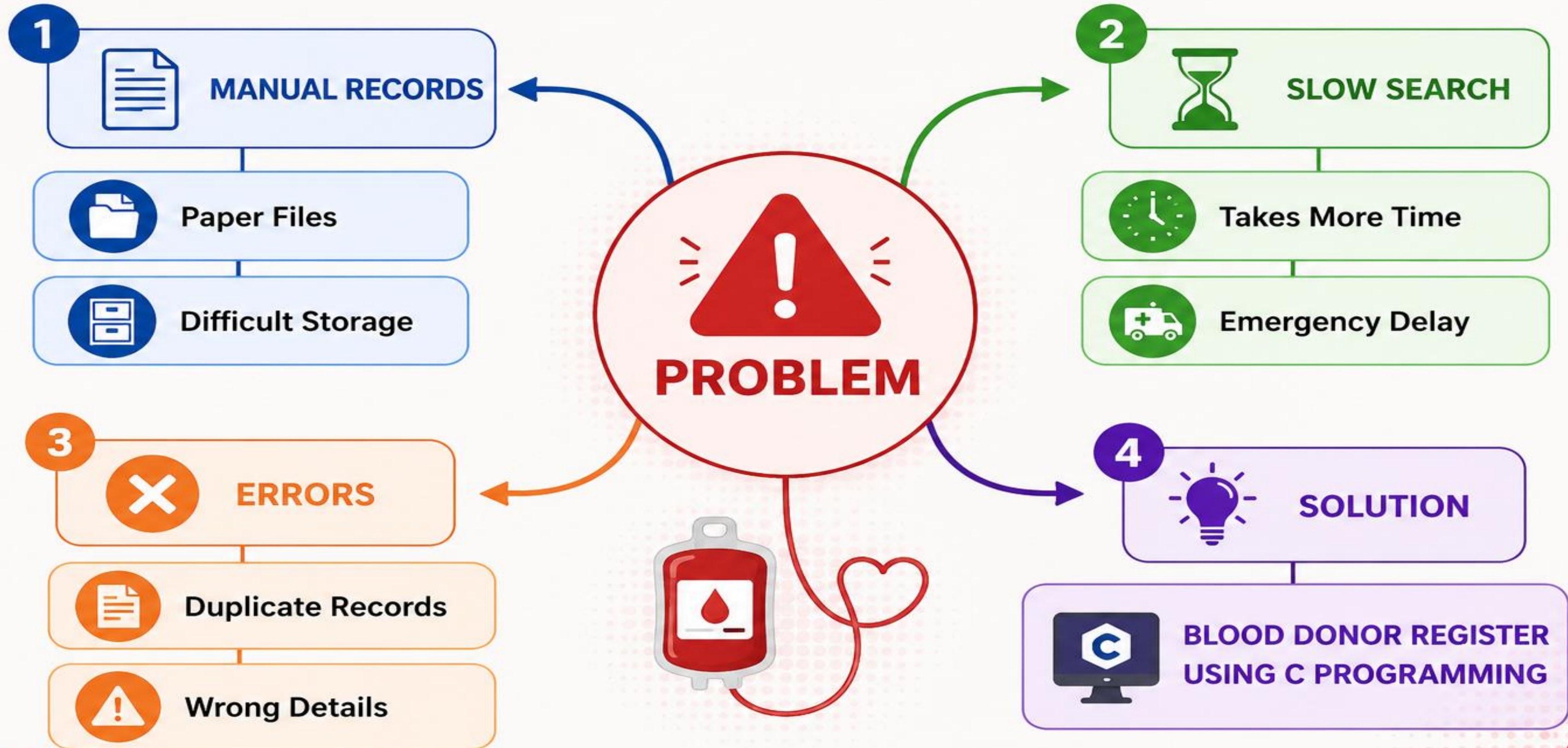
Introduction

-  Blood donation is a life-saving process that requires quick access to donor information.
-  Traditional paper-based records are time-consuming and difficult to maintain.
-  This project introduces a **Blood Donor Registry System** developed using **C Programming**.
-  The system digitally stores and manages donor details efficiently.
-  It enables fast registration, searching, updating, and deletion of donor records.
-  File handling ensures secure and permanent data storage.
-  The project demonstrates the practical application of **structures, functions, arrays, conditional statements, loops, and file handling** in C.
-  It provides a simple, menu-driven interface for easy and reliable donor management.

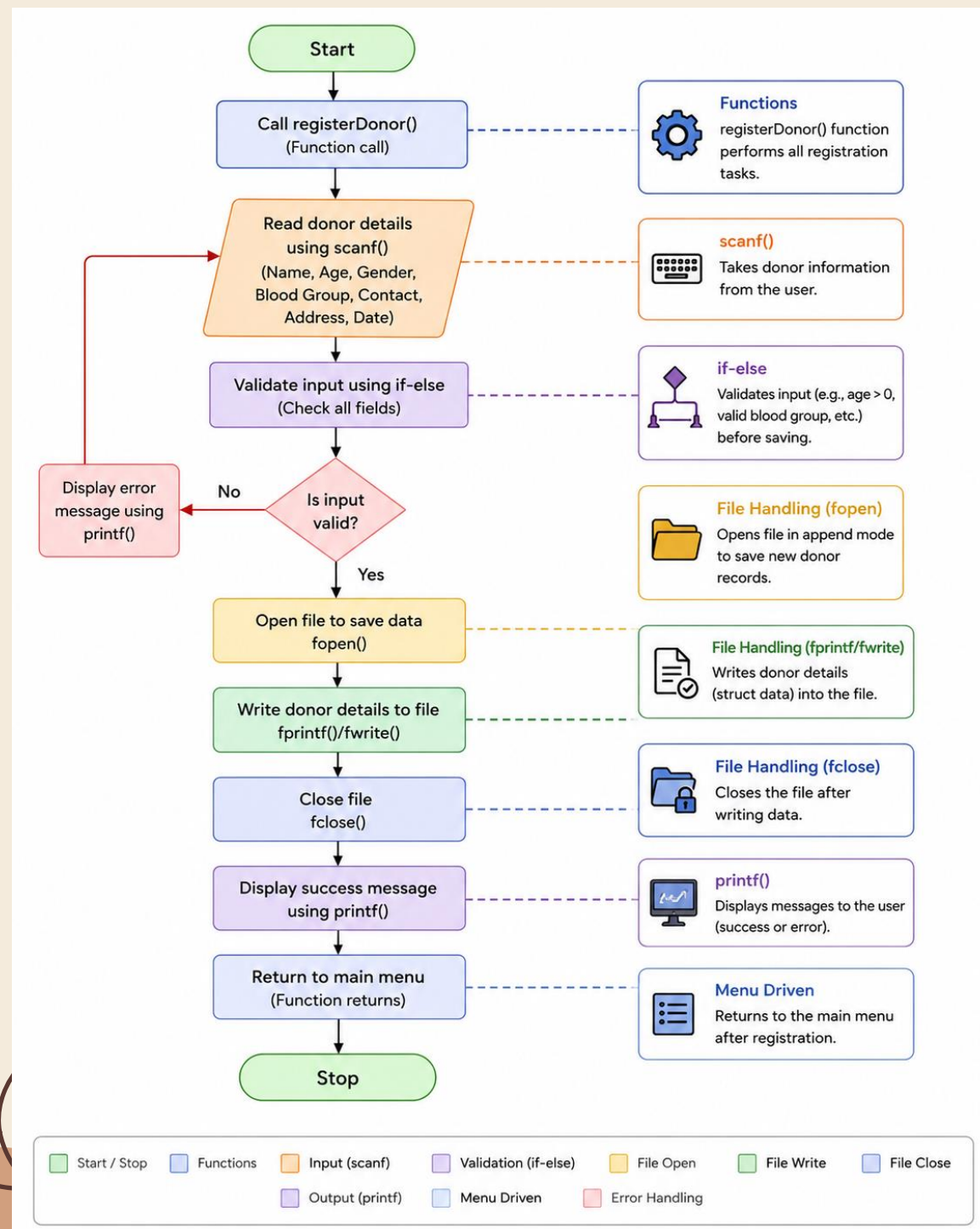




PROBLEM DERIVATION

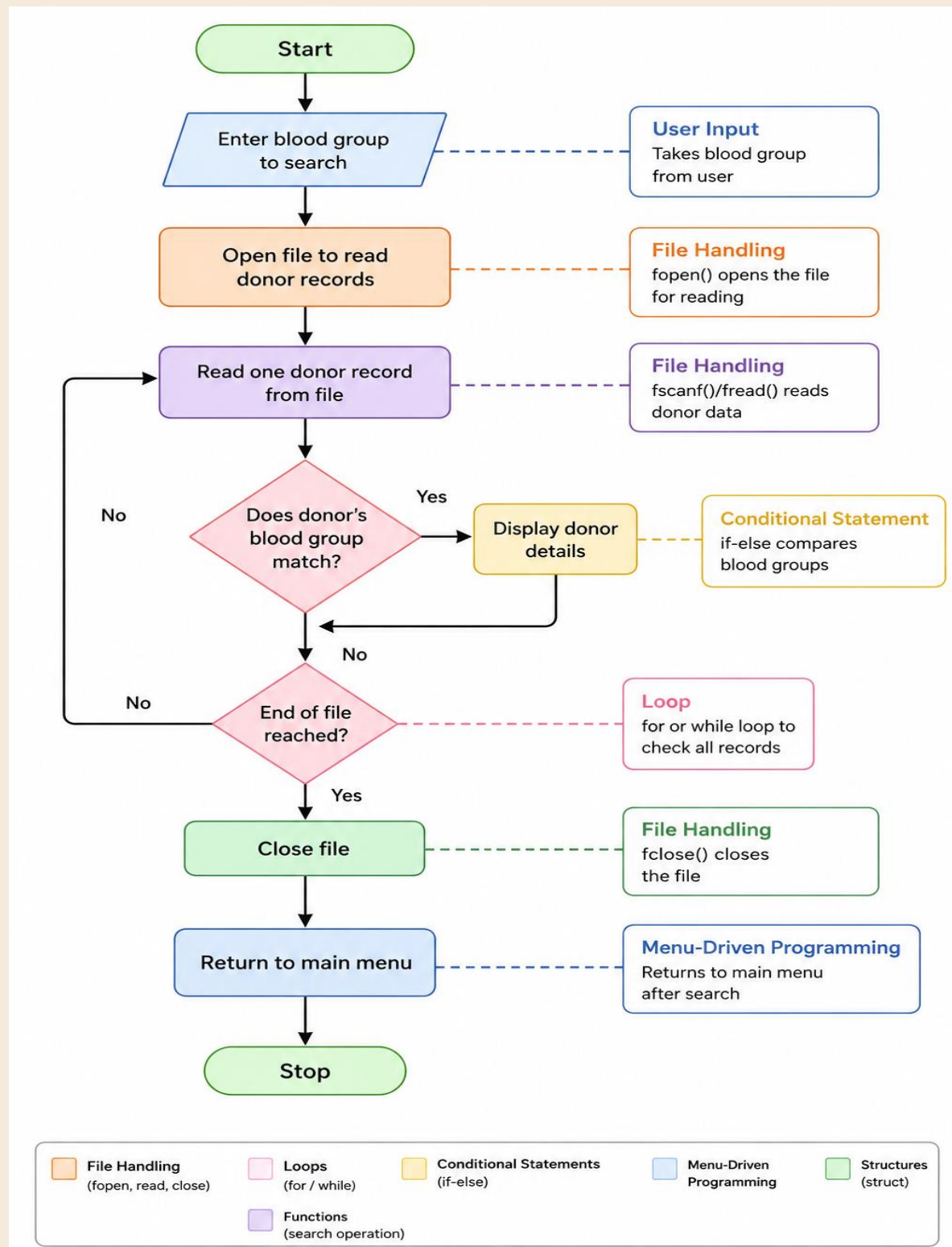


MODULE 1



- "Module 1 is the Donor Registration module.
- In the front end, the user enters the donor's name, age, gender, blood group, contact number, address, and registration date.
- The system validates the entered information before saving it.
- In the back end, C programming uses structures to store donor data, functions to process the input, and file handling to save the records permanently.
- After successful storage, the system confirms that the donor has been registered."

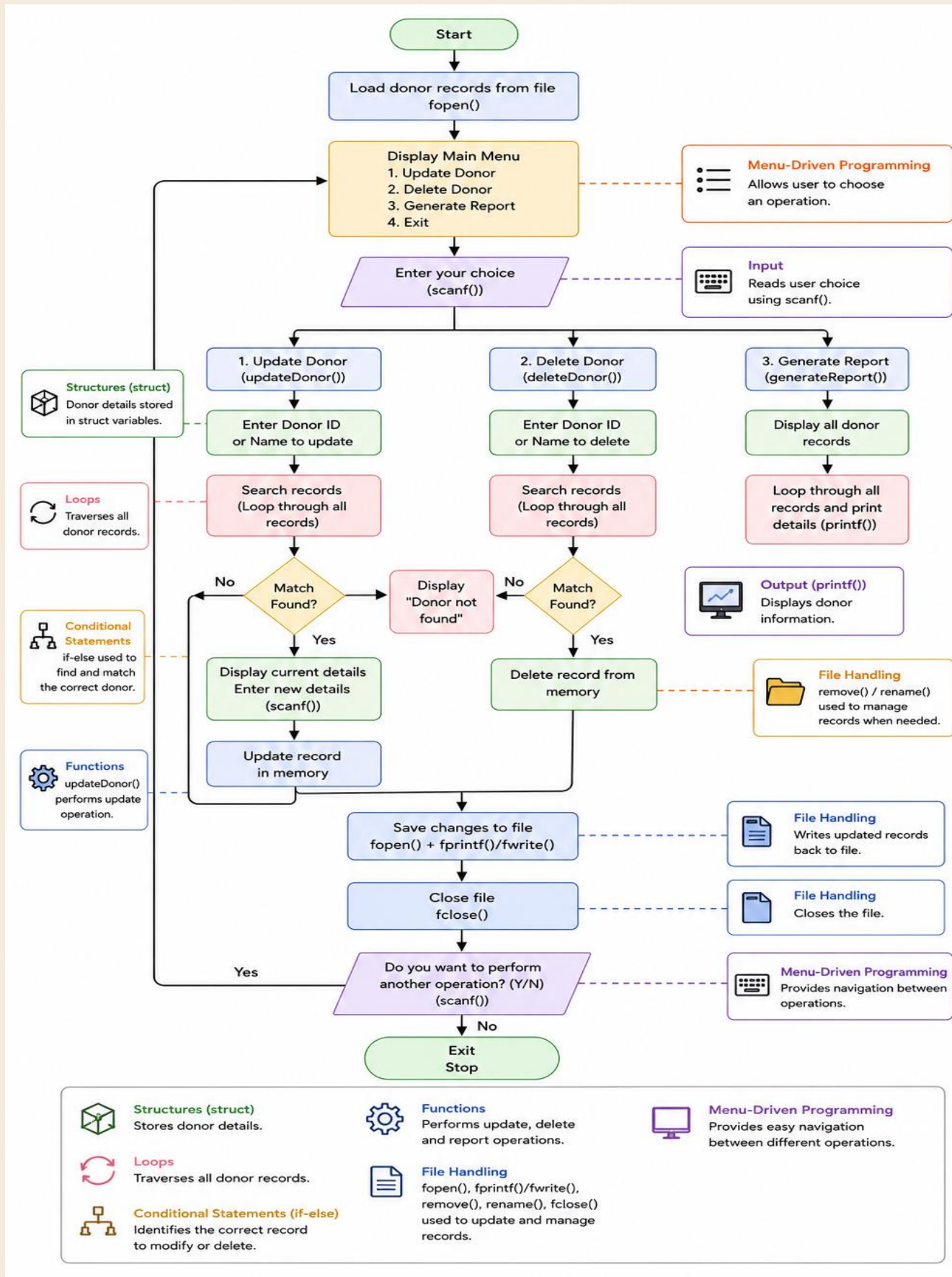
MODULE-2



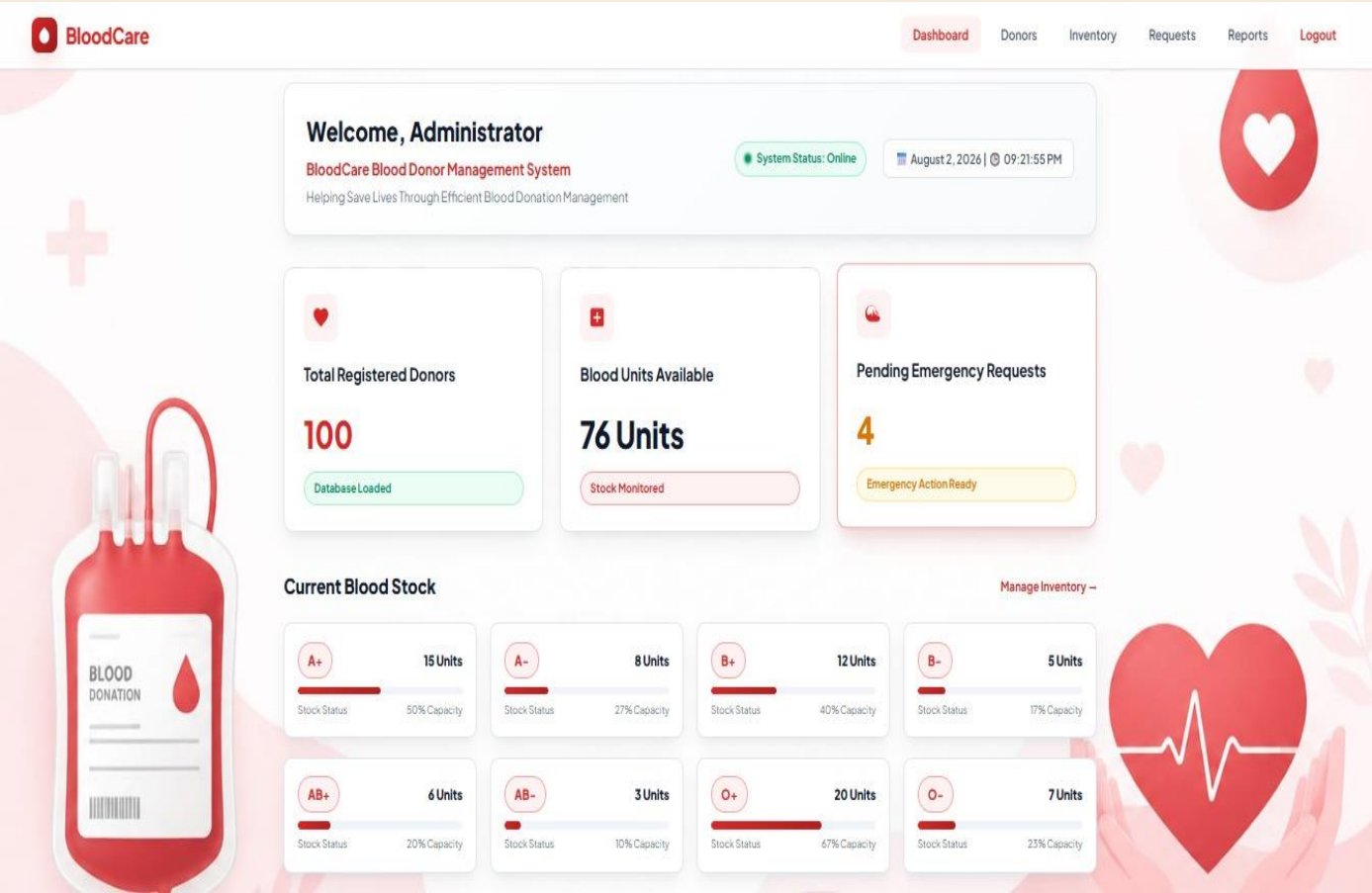
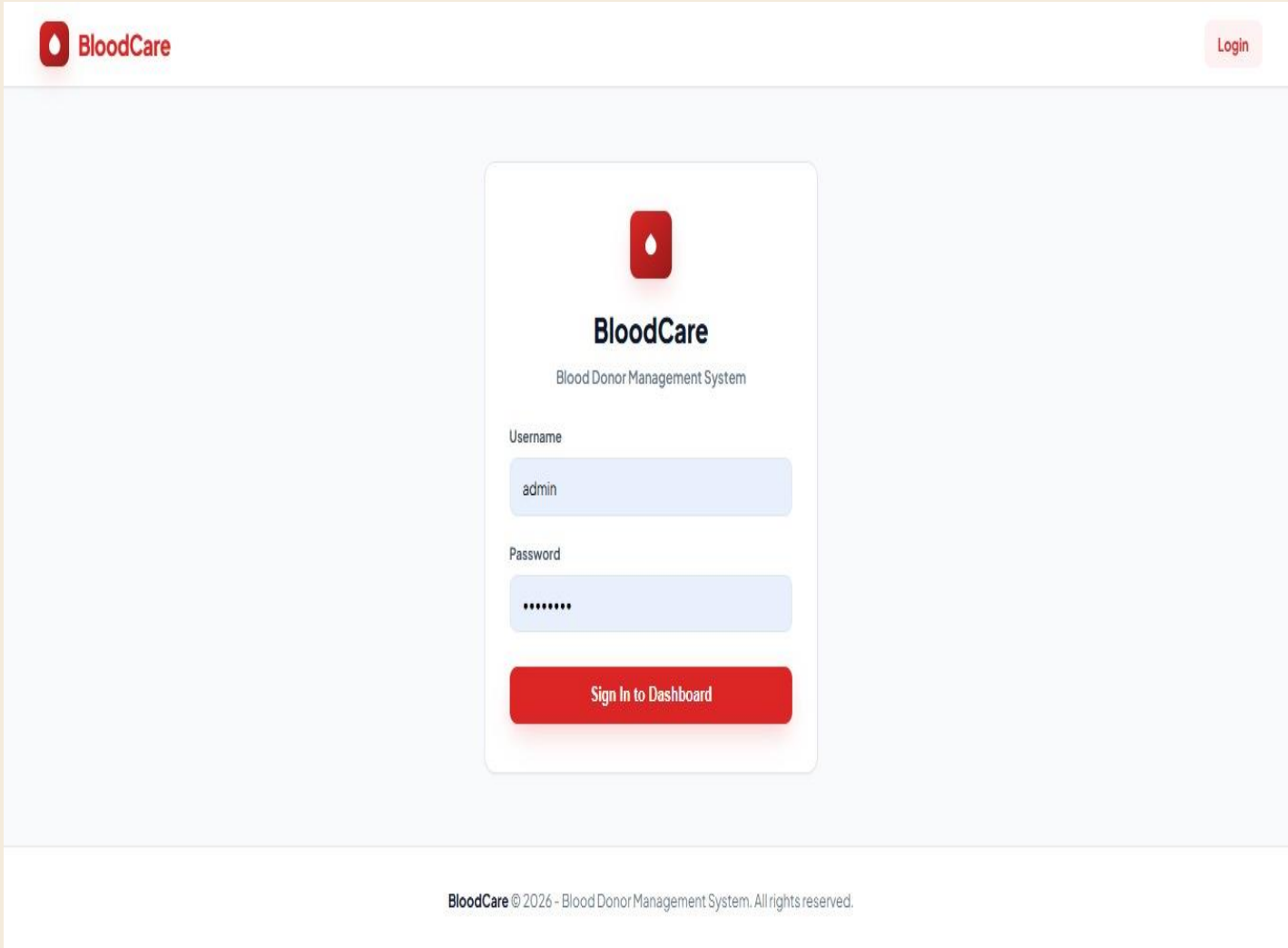
- "Module 2 is the Search Donor module. In the front end, the user selects the search option and enters the required blood group.
- The system searches the stored donor records and displays matching donor details.
- In the back end, C programming uses file handling to read records, loops to search through the data, and conditional statements to compare blood groups.
- If a match is found, the donor information is displayed; otherwise, the system shows that no matching donor is available."

MODULE-3

- "Module 3 is responsible for managing donor records."
- In the front end, the user can update donor information, delete records, view all donors, and generate reports.
- In the back end, C programming reads the existing records, modifies or deletes the selected record, stores the updated data in a temporary file, and replaces the old file.
- This module ensures that donor information remains accurate and up to date."



Result



BloodCare

Dashboard

Donors

Inventory

Requests

Reports

Logout

Emergency Blood Requests

Manage and fulfill urgent hospital & patient blood requests

+ Create New Request

Search requests by Patient Name, Hospital Name, or Blood Group...

4 Pending Requests

REQUEST ID	PATIENT NAME	HOSPITAL NAME	BLOOD GROUP	UNITS	CONTACT NUMBER	REQUEST DATE	STATUS	ACTIONS
REQ1002	Michael Brown	City Medical Centre	O+	1	+61 422333444	2026-08-03	Pending	Approve Reject Delete
REQ1003	Emily Davis	Central Hospital	B+	3	+61 433444555	2026-08-03	Approved	Delete

BloodCare

Dashboard

Donors

Inventory

Requests

Reports

Logout

Healthcare Reports & Analytics

Real-time analytics and statistics calculated by C backend from data files

Print / Save as PDF

Total Donors

100

Registered donor records

Total Blood Units

76

Units across 8 blood groups

Total Requests

7

Total hospital requests

Pending Requests

4

Action Required

Approved Requests

2

Fulfilled

Rejected Requests

1

Declined

BloodCare

Dashboard

Donors

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Blood Stock Inventory Management

Track and manage blood units across all 8 blood groups in real time

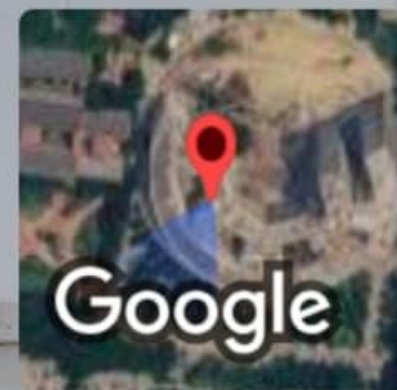
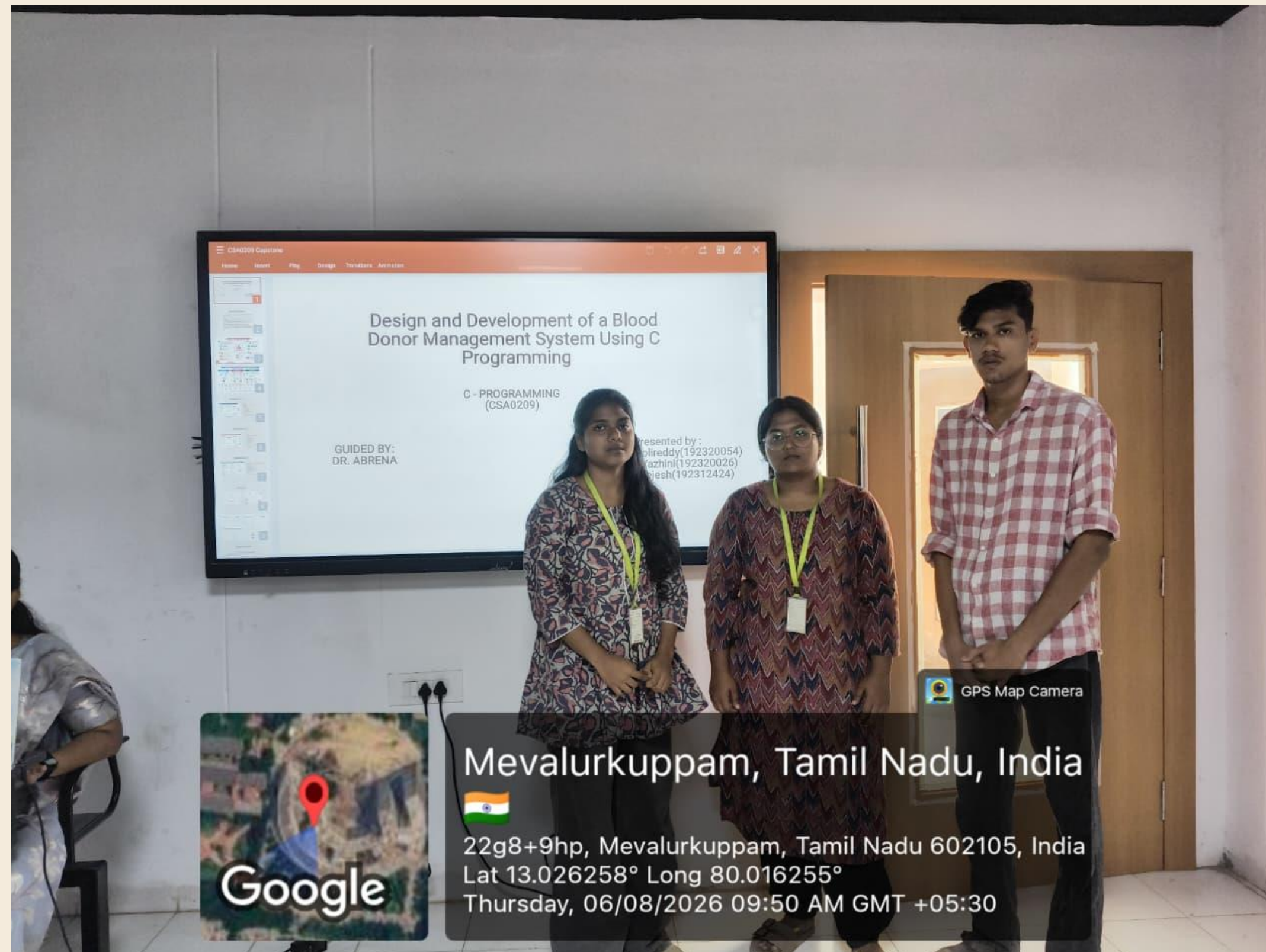
Search by blood group (e.g. A+, B-, O+)...

76 Total Units Available

BLOOD GROUP	UNITS AVAILABLE	LAST UPDATED	ACTIONS
A+	15 Units15 Units	2026-08-02 18:56:30	+ Add - Reduce Set Stock
A-	8 Units8 Units	2026-08-02 18:56:30	+ Add - Reduce Set Stock
B+	12 Units12 Units	2026-08-02 18:56:30	+ Add - Reduce Set Stock
B-	5 Units5 Units	2026-08-02 18:56:30	+ Add - Reduce Set Stock

Conclusion

- The Blood Donor Register simplifies the management of blood donor records.
- It is developed using C Programming with a menu-driven interface.
- The system uses structures, functions, and file handling to store and manage donor information.
- It enables quick donor registration, searching, updating, deleting, and report generation.
- It reduces manual paperwork, saves time, and improves data accuracy.
- This project demonstrates the practical application of C programming concepts in developing a real-world blood donor management system.



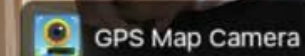
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Thank
You